PRACTICAL 1 - LINEAR (MIXED) MODELS

In this practical we are going to fit linear (mixed) models in `inlabru`. We are going to to:

- Fit a simple linear regression
- Fit a linear regression with discrete covariates and interactions
- Fit a linear mixed model.

> Note
>
> You can download the R-script of this practical by clicking the button below:

Start by loading useful libraries:

```
library(INLA)
library(patchwork)
library(inlabru)
library(car)
library(tidyverse)
# load some libraries to generate nice plots
library(scico)
```

At the end of this exercise we are going to compare the different models we fit using DIC and WAIC. Therefore we neet to tell the `bru()` function to compute those scores, we can do that by the `bru_options_set()` functions:

```
bru_options_set(control.compute = list(dic = T, waic = T))
```

This is a global option that will make `bru()` compute scores everytime. It is also possble to set the option locally as

```
fit = bru(cmp, lik,
          options = list(control.compute = list(dic = TRUE)))
```

1 Simple linear regression

We consider a simple linear regression model with Gaussian observations

$$y_i \sim \mathcal{N}(\mu_i, \sigma^2), \qquad i = 1, \dots, N$$

where σ^2 is the observation error, and the mean parameter μ_i is linked to the **linear predictor** (η_i) through an identity function:

$$\eta_i = \mu_i = \beta_0 + \beta_1 x_i.$$

Here $\mathbf{x} = (x_1, \dots, x_N)$ is a continuous covariate and β_0, β_1 are parameters to be estimated.

To finalize the Bayesian model we assign prior distribution as $\tau = 1/\sigma^2 \sim \text{Gamma}(a, b)$ and $\beta_0, \beta_1 \sim \mathcal{N}(0, 1/\tau_\beta)$ (we will use the default prior settings in INLA for now).

Question

What is the dimension of the hyperparameter vector and latent Gaussian field?
Answer
The hyperparameter vector has dimension 1, $\boldsymbol{\theta} = (\tau)$ while the latent Gaussian field $\boldsymbol{u} = (\beta_0, \beta_1)$ has dimension 2, 0 mean, and sparse precision matrix:

$$\boldsymbol{Q} = \begin{bmatrix} \tau_{\beta_0} & 0 \\ 0 & \tau_{\beta_1} \end{bmatrix}$$

Note that, since β_0 and β_1 are fixed effects, the precision parameters τ_{β_0} and τ_{β_1} are fixed.

Note

We can write the linear predictor vector $\boldsymbol{\eta} = (\eta_1, \dots, \eta_N)$ as

$$\boldsymbol{\eta} = \boldsymbol{A}\boldsymbol{u} = \boldsymbol{A}_1\boldsymbol{u}_1 + \boldsymbol{A}_2\boldsymbol{u}_2 = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \beta_0 + \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix} \beta_1$$

Our linear predictor consists then of two components: an intercept and a slope.

2 Modelling penguins bodymass

Let's look at the dataset `penguins` in `R`.

These are data on adult penguins covering three species found on three islands in the Palmer Archipelago, Antarctica, including their size (flipper length, body mass, bill dimensions), and sex.

```r
data("penguins")
glimpse(penguins)
```

```
Rows: 344
Columns: 8
$ species     <fct> Adelie, Adelie, Adelie, Adelie, Adelie, Adelie, Adelie, Ad~
$ island      <fct> Torgersen, Torgersen, Torgersen, Torgersen, Torgersen, Tor~
$ bill_len    <dbl> 39.1, 39.5, 40.3, NA, 36.7, 39.3, 38.9, 39.2, 34.1, 42.0, ~
$ bill_dep    <dbl> 18.7, 17.4, 18.0, NA, 19.3, 20.6, 17.8, 19.6, 18.1, 20.2, ~
$ flipper_len <int> 181, 186, 195, NA, 193, 190, 181, 195, 193, 190, 186, 180,~
$ body_mass   <int> 3750, 3800, 3250, NA, 3450, 3650, 3625, 4675, 3475, 4250, ~
$ sex         <fct> male, female, female, NA, female, male, female, male, NA, ~
$ year        <int> 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007~
```

The dataset contains the following variables:

species a factor denoting penguin species (Adélie, Chinstrap and Gentoo)

island a factor denoting island in Palmer Archipelago, Antarctica (Biscoe, Dream or Torgersen)

bill_length a number denoting bill length (millimeters)

bill_depth_mm a number denoting bill depth (millimeters)

flipper_length an integer denoting flipper length (millimeters)

body_mass an integer denoting body mass (grams)

sex a factor denoting penguin sex (female, male)

year and integer indicating the study year: (2007, 2008, 2009)

We are going to use the bodymass as our response variable. To make the interpretation easier we are going to express the bodymass in Kg instead of grams.

```
penguins$body_mass = penguins$body_mass/1000
```

2 Fitting a linear regression model with `inlabru`

In a first step we want to fit a simple linear regression model to link the body mass (`body_mass`) to flipper length (`flipper_len`).

$$y_i \sim \mathcal{N}(\mu_i, \sigma_y^2)$$
$$\mu_i = \beta_0 + \beta_i x_i$$

where y_i indicates the body mass and x_i the flipper length of penguin i.

Step1: Defining model components

The first step is to define the two model components: The intercept and the linear covariate effect.

Task

Define an object called `cmp` that includes and (i) intercept `beta_0` and (ii) and a linear effect `beta_1` of the `flipper_len`.
Take hint
The `cmp` object is here used to define model components. We can give them any useful names we like, in this case, `beta_0` and `beta_1`. You can remove the automatic intercept construction by adding a `-1` in the components
Click here to see the solution

```
cmp =  ~ -1 + beta_0(1) + beta_1(flipper_len, model = "linear")
```

Note

Note that we have excluded the default Intercept term in the model by typing `-1` in the model components. However, `inlabru` has automatic intercept that can be called by typing `Intercept()` , which is one of `inlabru` special names and it is used to define a global intercept, e.g.

```
cmp =  ~  Intercept(1) + beta_1(flipper_len, model = "linear")
```

another way to code this model would be to use the `model = "fixed"`. In this case we would define the components as:

```r
cmp = ~ -1 + effects(~ flipper_len, model = "fixed")
```

NOTE In the following we are going to use this last way of defining the components!

Step 2: Build the observation model

The next step is to construct the observation model by defining the model likelihood. The most important inputs here are the `formula`, the `family` and the `data`.

Task

Define a linear predictor `eta` using the component labels you have defined on the previous task.
Take hint
The `eta` object defines how the components should be combined in order to define the model predictor.
Click here to see the solution

```r
formula = body_mass ~ effects
```

The likelihood for the observational model is defined using the `bru_obs()` function.

Task

Define the observational model likelihood in an object called `lik` using the `bru_obs()` function.
Take hint
The `bru_obs` is expecting three arguments:

- The linear predictor `eta` we defined in the previous task
- The data likelihood (this can be specified by setting `family = "gaussian"`)
- The data set `df`

Click here to see the solution

```r
lik = bru_obs(formula = formula,
              data = penguins,
              family = "gaussian")
```

Step 3: Fit the model

We fit the model using the `bru()` functions which takes as input the components and the observation model:

```r
fit1 = bru(cmp, lik)
```

NOTE running the code above we are going to get an *error* !!

This error is liked to the fact that, when using `model = "fixed"` the default option for `model.matrix()` is to remove all lines with NAs.

NAs are not a problem in `inlabru` they just don't make any contribtion to the likelihood….but matrix of the wrong dimension are :smile:.

To avoid this we can tell `R` to keep all missing values as follows:

```
options(na.action = 'na.pass')
fit1 = bru(cmp, lik)
```

Step 4: Extract results

There are several ways to extract and examine the results of a fitted `inlabru` object.

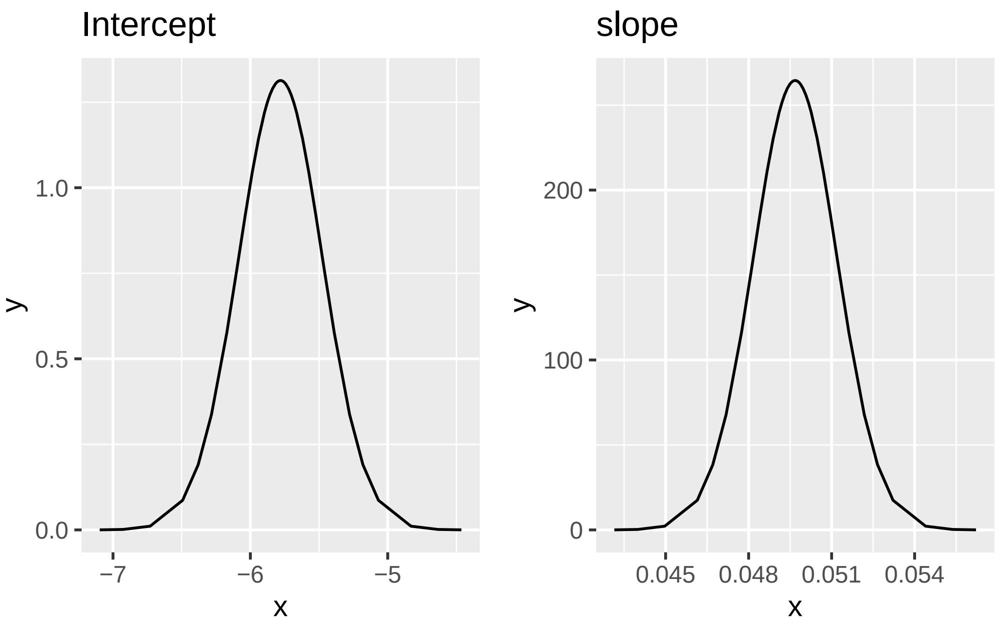
The most natural place to start is to use the `summary()` which gives access to some basic information about model fit and estimates

```
summary(fit1)
## inlabru version: 2.13.0.9032
## INLA version: 26.02.06
## Latent components:
## effects: main = fixed(~flipper_len)
## Observation models:
##   Model tag: <No tag>
##     Family: 'gaussian'
##     Data class: 'data.frame'
##     Response class: 'numeric'
##     Predictor: body_mass ~ effects
##     Additive/Linear/Rowwise: TRUE/TRUE/TRUE
##     Used components: effect[effects], latent[]
## Time used:
##     Pre = 0.537, Running = 0.191, Post = 0.0172, Total = 0.746
## Random effects:
##   Name     Model
##     effects IID model
##
## Model hyperparameters:
##                                         mean    sd 0.025quant 0.5quant
## Precision for the Gaussian observations 6.47 0.495       5.54     6.46
##                                         0.975quant mode
## Precision for the Gaussian observations       7.48 6.43
##
## Deviance Information Criterion (DIC) ...............: 337.94
## Deviance Information Criterion (DIC, saturated) ....: 347.38
## Effective number of parameters .....................: 2.99
##
## Watanabe-Akaike information criterion (WAIC) ...: 337.90
## Effective number of parameters .................: 2.92
##
## Marginal log-Likelihood:  -192.93
##  is computed
## Posterior summaries for the linear predictor and the fitted values are computed
## (Posterior marginals needs also 'control.compute=list(return.marginals.predictor=TRUE)')
```

Task

Plot the posterior marginals for the estimated intercept and slope
Click here to see the solution

```
p1 = fit1$marginals.random$effects$index.1 %>% ggplot() + geom_line(aes(x,y)) + ggtitle("I
p2 = fit1$marginals.random$effects$index.2 %>% ggplot() + geom_line(aes(x,y)) + ggtitle("s
p1 + p2
```



Another way, which gives access to more complicated (and useful) output is to use the `predict()` function.

Below we take the fitted `bru` object and use the `predict()` function to produce predictions for η given a new set of values for the years of service

```
new_data = data.frame(flipper_len  = 170:240)
pred = predict(fit1, new_data, ~ effects,
               n.samples = 1000)
```

The `predict()` function generate samples from the fitted model and then summarizes those samples by computing some statistics like posterior mean, standard deviation, quantiles etc. In this case we set the number of samples to 1000.

We can plot the predictions for η together with the observed data:

3 Plot

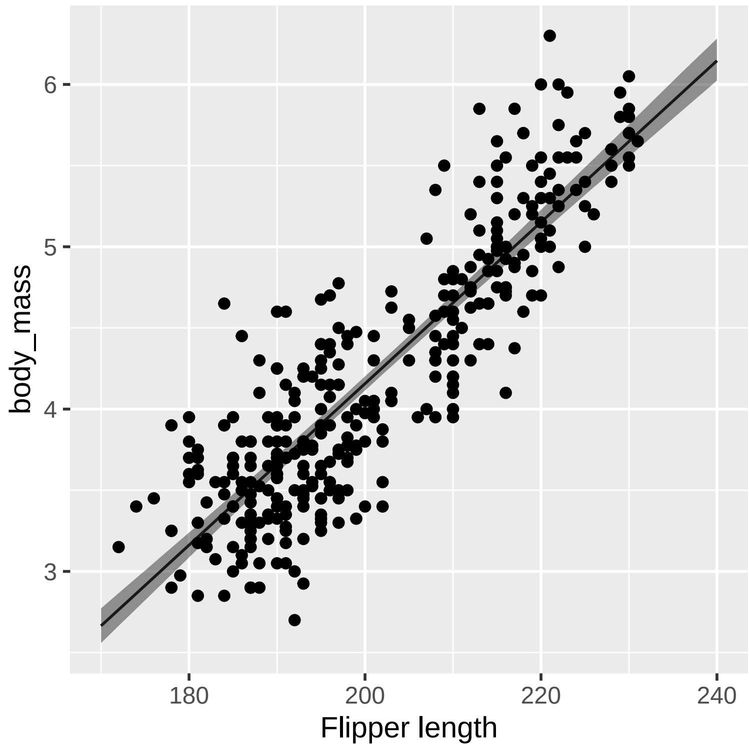


Figure 1: Data and 95% credible intervals

NOTE: The uncertainty we have computed now is relative to the prediction of η. If we want to predict new body mass, we need to add the observation (likelihood) uncertainty σ_y^2. We can do this using `predict()` again:

```
new_data = data.frame(flipper_len  = 170:240)

pred1 = predict(fit1, new_data,
               formula = ~ { eta = effects
                             sigma = sqrt(1/Precision_for_the_Gaussian_observations)
                             list(mean = eta,
                                  q1 = qnorm(0.025, mean = eta, sd = sigma),
                                  q2 =  qnorm(0.975, mean = eta, sd = sigma))
                            },
               n.samples = 1000)
```

Let's compare the two predictions:

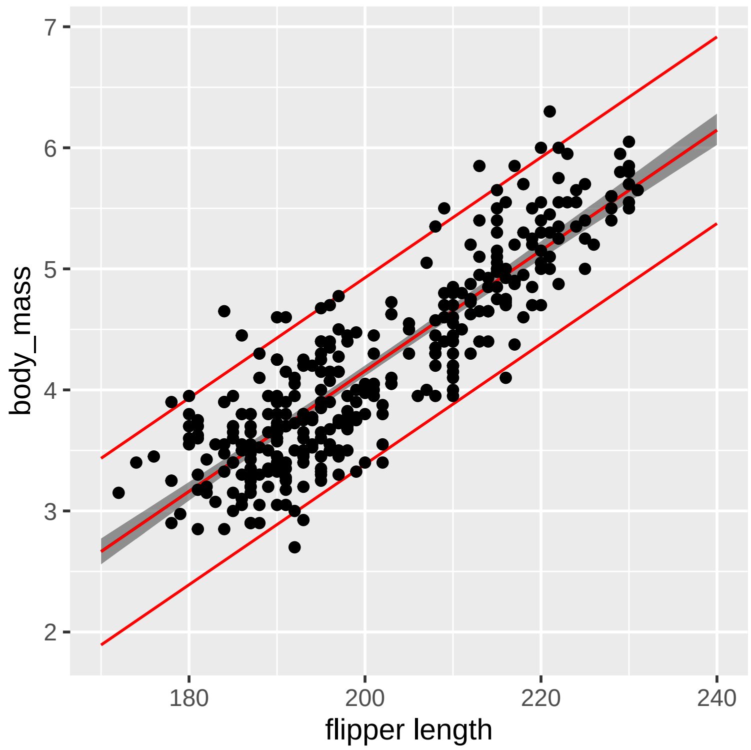


Figure 2: Data and 95% credible intervals

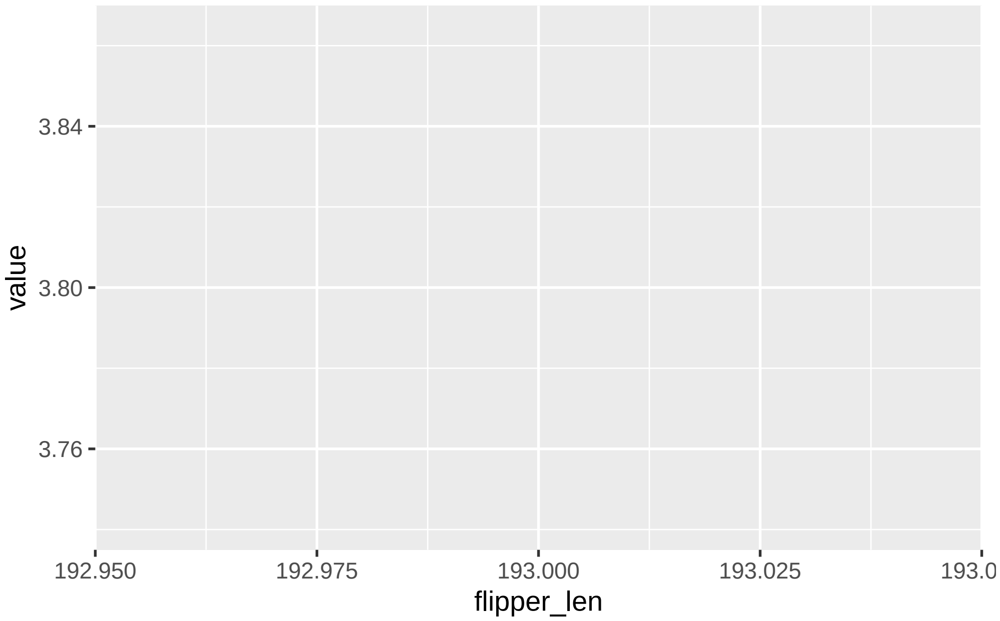
Task

Use the function generate() to create a sample of 100 possible regression lines from the fitted model
Click here to see the solution

```
new_data = data.frame(flipper_len = 193)
gen = generate(fit1, new_data, ~ effects,
               n.samples = 100)

data.frame(flipper_len = new_data$flipper_len,
           gen) %>%
  pivot_longer(-flipper_len) %>%
  ggplot() + geom_line(aes(flipper_len, value, group = name))
```

```
`geom_line()`: Each group consists of only one observation.
i Do you need to adjust the group aesthetic?
```



Task

Generate predictions for the mean body mass η when the flipper length is $x_0 = 193$ mm

What is the predicted value for η? And what is the uncertainty?

Take hint

You can create a new data frame containing the new observation x_0 and then use the `predict` function.

Click here to see the solution

```
new_data = data.frame(flipper_len = 193)
pred = predict(fit1, new_data, ~ effects,
                n.samples = 1000)

pred
```

```
  flipper_len     mean         sd    q0.025     q0.5   q0.975    median
1         193 3.808332 0.02402348 3.762607 3.808033 3.853782 3.808033
  mean.mc_std_err sd.mc_std_err
1    0.0007938817    0.00054063
```

You can see the predicted mean and sd by examining the produced `pred` object. In this case the mean is ca 4 kg with sd ca 0.02. This gives a 95% CI ca [3.76, 3.85].

NOTE Now we have produced a credible interval for the expected mean η if we want to produce a *prediction* interval for a new observation y we need to add the uncertainty that comes from the likelihood with precision $\tau = 1/\sigma_y^2$. To do this we can again use the `predict()` function to compute a 95% prediction interval for y.

```
pred2 = predict(fit1, new_data,
               formula = ~ {
                 mu = effects
                 sigma = sqrt(1/Precision_for_the_Gaussian_observations)
                 list(q1 = qnorm(0.025, mean = mu, sd = sigma),
```

```
                        q2 =  qnorm(0.975, mean = mu, sd = sigma))},
                 n.samples = 1000)
round(c(pred2$q1$mean, pred2$q2$mean),2)
```

```
[1] 3.04 4.58
```

Notice that now the interval we obtain is much bigger! ahahah

4 Linear model with discrete variables and interactions

Now we want to check if there is any difference, both in mean body_mass and in mean increase of the body_mass with flipper length, between males and females

To do this we have to fit a model with interactions:

$$
\begin{aligned}
y_i &\sim \mathcal{N}(\eta_i, \sigma_y^2) \\
\eta_i &= \beta_0 + \beta_{0,\text{Male}} + \beta_1 x_i + \beta_{1,\text{Male}} x_i
\end{aligned}
$$

Let's first look at an exploratuve plot

```
penguins %>%
  filter(!is.na(sex)) %>%
  ggplot() + geom_point(aes(flipper_len, body_mass, color= sex)) +
  facet_wrap(.~sex)
```

We fit the model using the `model = "fixed"` option.

```
cmp = ~ -1 + effects( ~ sex*flipper_len, model = "fixed")
formula = body_mass ~ .
lik = bru_obs(formula = formula,
```

```
                 data = penguins
                 )
fit2 = bru(cmp, lik)
```

We can get the results looking at the `summary.random` object as follows:

```
fit2$summary.random
```

```
$effects
                   ID          mean          sd    0.025quant       0.5quant
1         (Intercept) -5.2560419315 0.423683127 -6.087343982 -5.2560447990
2             sexmale  0.2188495010 0.575523540 -0.910395725  0.2188524648
3         flipper_len  0.0462183314 0.002141407  0.042016627  0.0462183459
4 sexmale:flipper_len  0.0006403378 0.002862713 -0.004976558  0.0006403229
    0.975quant          mode          kld
1 -4.424723521 -5.2560447955 5.442482e-10
2  1.348077817  0.2188524611 5.440492e-10
3  0.050419953  0.0462183459 5.442672e-10
4  0.006257319  0.0006403229 5.440460e-10
```

Task

Use the `predict()` function to get an estimate of the mean increase of body_mass per mm of flipper length for males and females.
Would you conclude that there are differences between males and females?
Give me a hint
Since we are not interested in prediction but just on the parameters values, we can use the `_latent` suffix to get only the estimated parameters
Click here to see the solution

```
# Here we use the _latent "trick" to recovere the parameters
# here we do not need any new data to predict for, we can use an
# empty data frame

params = predict(fit2, data.frame() , ~ data.frame( intercept_female = effects_latent[1],
                                          intercept_male = effects_latent[1] + effects_
                                          slope_female = effects_latent[3]  ,
                                          slope_male = effects_latent[3] + effects_late
```

Two things to notice here:

1. We have used a empty object as `newdata` in the `predict()` function.
2. The suffix `_latent` indicates that we are interested in the latent model efffect.

Task

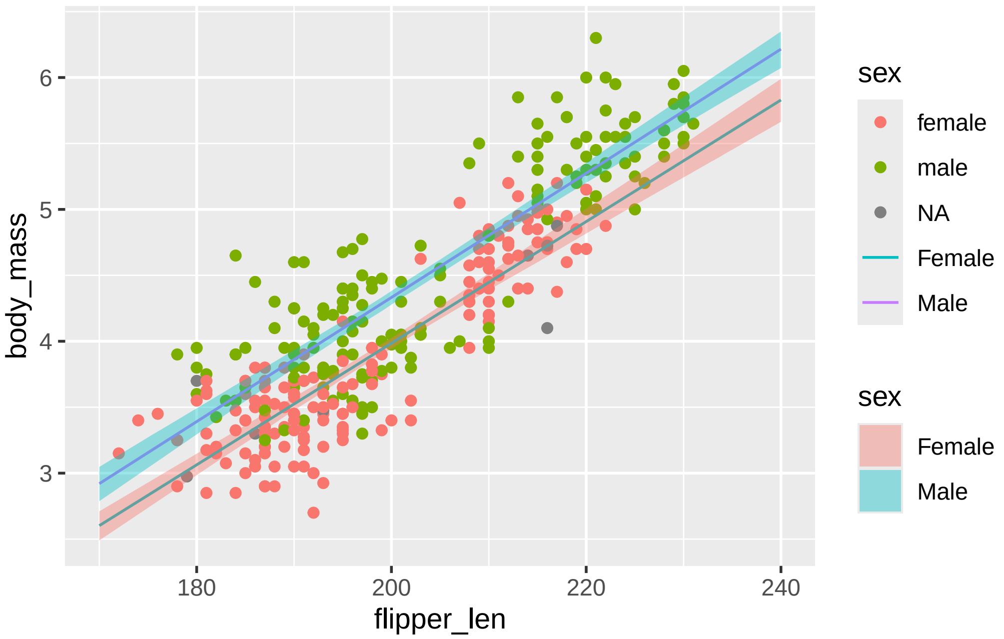
Use the `predict()` function again, this time to obtain the two regression lines.
Click here to see the solution

```r
new_data = data.frame(flipper_len = rep(170:240, 2),
                      sex = rep(c("Male", "Female"), each = 71))

pred = predict(fit2, new_data , ~ effects)


ggplot() + geom_point(data = penguins, aes(flipper_len, body_mass, color = sex)) +
  geom_line(data = pred, aes(flipper_len, mean, color = sex, group = sex)) +
  geom_ribbon(data = pred, aes(flipper_len, ymin = q0.025, ymax = q0.975, fill = sex, grou
              alpha = 0.4)
```

```
Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Another way to fit this model is to realize that a fixed effect can be seen as an iid effect with fixed precision, one of the `inlabru` ways to fit the model is:

```r
cmp = ~ -1 + sex_intercept(sex, model = "iid", initial = log(0.001), fixed = T) +
  sex_slope(sex, flipper_len,  model = "iid", fixed = T, initial = log(0.001))
lik = bru_obs(formula = body_mass ~ .,
              data = penguins %>% filter(!is.na(sex)))
fit2b = bru(cmp, lik)
```

Notice that we fix the precision of the `iid` effect to the same value as the precision of the linear effects which is 0.001.

The fitted values can be inspected as

```r
fit2b$summary.random$sex_intercept
```

```
      ID      mean        sd 0.025quant  0.5quant 0.975quant      mode
1 female -5.442907 0.4399821  -6.306203 -5.442910  -4.579593 -5.442910
2   male -5.036399 0.3884272  -5.798540 -5.036401  -4.274245 -5.036401
```

```
          kld
1 5.835819e-10
2 5.837102e-10
```

```
fit2b$summary.random$sex_slope
```

```
      ID       mean          sd 0.025quant   0.5quant 0.975quant       mode
1 female 0.04714741 0.002224866 0.04278187 0.04714742 0.05151285 0.04714742
2   male 0.04685481 0.001894586 0.04313734 0.04685482 0.05057221 0.04685482
          kld
1 5.835703e-10
2 5.836619e-10
```

The results from the `fit2b` model is not the same we get from the `fit` model. What is happening? It is just a matter of parametrization.

You can go from one parametrization to the other by using the `predict()` function as we saw earlier

5 Linear Mixed Model

When looking at the data we see that there are differences in body mass depending on the species.

To account for this we introduce `species` as a random effect in the model.

Task

Modify the code below to include an iid random effect for species:

```
cmp = ~ -1 + effects( ~ sex*flipper_len, model = "fixed") +
  species(species, ... )
formula = ...
lik = bru_obs(formula = formula,
              data = penguins
              )
fit3 = bru(cmp, lik)
```

Click here to see the solution

```
cmp = ~ -1 + effects( ~ sex*flipper_len, model = "fixed") +
  species(species, model = "iid")
formula = body_mass ~ .
lik = bru_obs(formula = formula,
              data = penguins
              )
fit3 = bru(cmp, lik)
```

We now look at the results for the fixed effecs and compare with those from the previous model

```
fit3$summary.random$effects[,c(1,3,5)]
```

```
                   ID          sd     0.5quant
1         (Intercept) 0.719605428  0.421091137
2             sexmale 0.482491529 -0.475715368
3         flipper_len 0.003370553  0.017276727
4 sexmale:flipper_len 0.002412976  0.005011396
```

```
fit2$summary.random$effects[,c(1,3,5)]
```

```
                   ID          sd      0.5quant
1         (Intercept) 0.423683127 -5.2560447990
2             sexmale 0.575523540  0.2188524648
3         flipper_len 0.002141407  0.0462183459
4 sexmale:flipper_len 0.002862713  0.0006403229
```

Task

We want to include a random effect, dependent on species, also on the slope. Modify the code below to include that random effect.

```r
cmp = ~ -1 + effects( ~ sex*flipper_len, model = "fixed") +
  species1(species, ... ) +
  species2(species, ... )
formula = ...
lik = bru_obs(formula = formula,
              data = penguins
              )
fit4 = bru(cmp, lik)
```

Click here to see the solution

```r
cmp = ~ -1 + effects( ~ sex*flipper_len, model = "fixed") +
  species1(species, model = "iid") +
  species2(species, flipper_len, model = "iid")
formula = body_mass ~ .
lik = bru_obs(formula = formula,
              data = penguins
              )
fit4 = bru(cmp, lik)
```

6 Model comparison

As a by-product of the main computations `inlabru` can compute some scores that can be useful to compare models (we will talk more about this later in the course).

The easiest scores are the Deviance Information Criteria (DIC) and the Wakaike Information Criteria (WAIC).

These scores take into account goodness-of-fit and a penalty term that is based on the complexity of the model via the estimated effective number of parameters.

The lower the value of the score, the better the model is rated.

```r
deltaIC(fit1, fit2, fit3, fit4, criterion = c("DIC","WAIC"))
```

```
  Model      DIC  Delta.DIC     WAIC Delta.WAIC
1  fit4 147.3482   0.000000 147.1491   0.000000
2  fit3 148.4950   1.146744 148.3034   1.154309
3  fit2 272.9029 125.554681 272.6083 125.459207
4  fit1 337.9389 190.590663 337.8981 190.749073
```